

Clist_fix_operate.c3.65 KIB

```
1  #include <stdio.h>
2  #include <stdlib.h>
3  #include <string.h>
4
5  typedef struct _elem{
6      char name[21];
7      int age;
8      int num;
9  } Elem;
10
11 typedef struct _list
12 {
13     struct _list *prev;
14     struct _list *next;
15     Elem *data;
16 } List;
17
18 List *head, *tail;
19
20 int insert_one_list(char* input_name, int input_age, int input_num)
21 {
22     List *dummy;
23     Elem *content;
24
25     List *curr_target;
26
27     dummy = (List*)malloc(sizeof(List));
28     memset(dummy, 0, sizeof(List));
29
30     content = (Elem*)malloc(sizeof(Elem));
31     memset(content, 0, sizeof(content));
32     strncpy(content->name, input_name, sizeof(char) * 21);
33     content->age = input_age;
34     content->num = input_num;
35     dummy->data = content;
36
37     if(head)
38     {
39         curr_target = head; //not head->next
40     }else
41     {
42         return 0;
43     }
44
45     do{
46         if(tail != curr_target->next)
47         {
48             curr_target = curr_target->next;
49             continue;
50         }
51         else
52         {
53             dummy->prev = curr_target; //need treat dummy
54             dummy->next = tail; //need treat dummy
55             curr_target->next = dummy;
56             tail->prev = dummy;
57             break;
58         }
59     }while(1);
60
61     return 0;
62 }
63
64 int print_all_list()
65 {
66     List *curr_target;
67
68     if(head)
69     {
70         curr_target = head; //not head->next
71     }else
72     {
73         return 0;
74     }
75
76     do{
77         if(tail != curr_target->next)
78         {
79             curr_target = curr_target->next;
80             if(curr_target->data)
81             {
82                 printf("display: (%s) (%d) (%d)\n"
83                     , curr_target->data->name, curr_target->data->age, curr_target->data->num);
84             }
85             continue;
86         }
87         else
88         {
89             break;
90         }
91     }while(1);
92
93     return 0;
94 }
95
96 List* create_head_tail()
97 {
98     List *dummy;
99     dummy = (List*)malloc(sizeof(List));
100    memset(dummy, 0, sizeof(List));
101
102    return dummy;
103 }
104
105 int release_all_list()
106 {
107     List *curr_target, *prev_target;
108
109     int i = 0;
110
111     if(head)
112     {
113         curr_target = head;
114     }
115
116     do{
117         if(tail != curr_target->next)
118         {
119             printf("exist %d\n", ++i);
120             prev_target = curr_target;
121             curr_target = curr_target->next; // prior execute cause effect of (1)
122             if(prev_target->data)
123             {
124                 free(prev_target->data);
125             }
126             free(prev_target); // (1) changed ->next after free
127             continue;
128         }
129         else{
130             if(curr_target->data)
131             {
132                 free(curr_target->data);
133             }
134             free(curr_target);
135
136             free(tail);
137             printf("reach end !!!\n");
138             break;
139         }
140     }while(1);
141
142     return 0;
143 }
144
145 #if 0
146 int release_all_list()
147 {
148     List *curr_target, *prev_target;
149
150     int i = 0;
151
152     if(head)
153     {
154         curr_target = head->next;
155         free(head);
156     }
157
158     do{
159         if(!curr_target)
160         {
161             break;
162         }
163         else
164         {
165             if(tail != curr_target)
166             {
167                 printf("exist %d\n", ++i);
168             }
169             else{
170                 printf("tail !!!\n");
171             }
172             prev_target = curr_target;
173             curr_target = curr_target->next;
174             if(prev_target->data)
175             {
176                 free(prev_target->data);
177             }
178             free(prev_target);
179         }
180     }while(1);
181
182     return 0;
183 }
184 #endif
185
186 int main(int argc, char* argv[])
187 {
188
189     head = create_head_tail();
190     tail = create_head_tail();
191
192     head->next = tail;
193     tail->prev = head;
194
195     insert_one_list("dog", 10, 1);
196     insert_one_list("cat", 13, 2);
197
198     print_all_list();
199
200     release_all_list();
201
202     return 0;
203 }
204
205
```